

Dissecting the software-based measurement of CPU energy consumption: a comparative analysis

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Abstract—Every day, we experience the effects of the global warming: extreme weather events, major forest fires, storms, global warming, etc. The scientific community acknowledges that this crisis is a consequence of human activities where Information and Communications Technologies (ICT) are an increasingly important contributor.

Computer scientists need tools for measuring the footprint of the code they produce. Running Average Power Limit (RAPL) is a low-level interface designed by Intel that provides a measure of the energy consumption of a CPU (and more) without the need for additional hardware. Since 2017, it is available on most computing devices, including non-Intel devices such as AMD processors. More and more people are using RAPL for energy measurement, mostly like a black box without deep knowledge of its behaviour.

In this paper, we propose to come back to the basic mechanisms that allow to use RAPL measurements and present a critical analysis of their operations. For each mechanism, we release a reference implementation in Rust that avoids the pitfalls we detected in existing tools, improving correctness, timing accuracy and performance. In addition to long-established methods, we explore the suitability of the recent eBPF technology for working with RAPL.

We also provide an experimental study with multiple benchmarks and processor models in order to evaluate the efficiency of the various mechanisms and their impact on parallel software. Our experiments show that no mechanism provides a significant performance advantage over the others. However, they differ significantly in terms of ease-of-use and resiliency.

We believe that this work will help the community to develop correct, resilient and lightweight measurement tools, based on the mechanism that suits their needs.

Index Terms—energy consumption, energy efficiency, performance analysis, software measurement, RAPL library (Running Average Power Limit)

I. INTRODUCTION

A. Context and motivation

The impact of ICT in the environmental crisis has unfortunately increased and it is likely going to continue increasing [1, 2, 3, 4, 5]. The carbon footprint of the field was evaluated to about 1.2 to 2.2 Giga tons equivalent CO₂ (denoted in short by GtCO₂e) in 2020, which corresponds to about 2.1 to 3.9% of worldwide greenhouse gases (GHG) emission [6]. They are expected to reach 5.1 to 5.3 GtCO₂e by 2040 [3]. Yet, ICT are also often presented as an effective solution for decreasing the environmental footprint of other sectors, mostly thanks to optimization and substitution effects [7, 8]. Some reports hence claim that ICT's environmental benefits

can be several times greater than their own environmental burden [9]. Nevertheless, the assessment of these benefits generates a lot of controversy [6, 10, 11], as the global pressure of humanity on the environment keeps on increasing [12, 13]. In this context, measuring the energy consumption of ICT is required to become aware of its impact and establish environmentally friendly practices such as optimization efforts, power capping, restriction use, etc. In particular, knowing the energy consumption of a system, be it a whole supercomputer platform, a single node or a particular piece of software, allows to estimate its carbon footprint.

Frontier was the first high performance computing (HPC) platform to reach the real exascale in june 2023 [14], it needs 21.1 Megawatts of electricity, which generates 150 tons of CO₂e emissions a day. With such an impact, HPC systems are more and more studied from the perspective of energy efficiency. Since 2013, the supercomputers are ranked by energy efficiency in the Green500 [15]. Section 4 of the paper of Khan et al. [16] shows that energy efficiency is increasing in the Top500, but at a lower rate as computing performance. Assessing the energy efficiency calls for measuring the energy consumption.

There are multiple families of tools that can be used to collect the energy consumption ranging from physical wattmeters (e.g. connected PDUs – Power Distribution Units – or BMCs – Baseboard Management Controllers) to web browser extensions that run estimation models. They have different accuracy, acquisition frequency and ease-of-use.

In this paper, we focus on the Running Average Power Limit (RAPL) technology [17], which is the basic building block for many software measurement tools. RAPL measures the electricity consumption of the CPU and more components thanks to sensors integrated into the system-on-chip, and exposes it to the operating system through model-specific registers (MSR in short) designed by the CPU manufacturer. The advantage of RAPL is that no external powermeter is required, nor a privileged access to the BMC (which could be used to power off the server). It also provides more details than a PDU or a BMC, since it monitors internal components of the node. Moreover, RAPL is more accurate than any statistical estimation model, even though they can be tuned to reduce their error [18, 19]. The underlying philosophy of software measurement tools based on RAPL is that each device owner monitors the software running on it.

Extracting the measurements from the CPU requires an

interface. One can either use the low-level RAPL MSR directly, or choose a higher-level interface provided by the operating system. Linux provides two of them, namely, the Power Capping framework (powercap) and the Performance Counters subsystem (perf-events). These software interfaces query the processor registers provided by RAPL.

Sometimes, the same interface can be used in a number of very different ways. This is the case with perf-events, which can be read from user space or from kernel space using eBPF. We propose to study these two ways separately. This makes a total of four mechanisms that can be chosen to extract the consumption measurements from the CPU: MSR, powercap, perf-events in user space, and perf-events with eBPF. They are described in section III.

Higher level software measurement tools, such as CodeCarbon [20], PowerAPI [21] and Scaphandre [22], are based on these raw mechanisms. They use RAPL measurements to estimate the electricity consumption of each active process.

B. Objectives

The purpose of this work is threefold:

- provide RAPL users with a deep understanding of its access mechanisms and of the associated best practices.
- highlight the qualitative differences between the measurement mechanisms, based on criteria such as required expertise level and resiliency. This highlights the features and trade-offs of the mechanisms.
- evaluate the overhead of the various mechanisms. In particular, we address the following questions:
What is the overhead on the other programs of measuring the energy consumption of the CPU? What is the impact of the measurement on an idle machine? Are some mechanisms more efficient than others? Is there a difference between Intel and AMD processors?

To the best of our knowledge, these issues have not been fully addressed yet.

C. Contributions

To achieve the previous objectives, we propose a minimal measurement tool that includes an implementation of each mechanism. We release it with an open-source license in order to provide a reference implementation to the community. We test different strategies to avoid the common mistakes that we detected in other measurement tools. Furthermore, we analyze and compare the four measurement mechanisms.

Using this minimal tool, we experimentally study the impact of the measurement on parallel software and on an idle server. Each mechanism has been benchmarked with several HPC testbeds, CPU vendors and RAPL domains. The benchmark results have been accumulated during approximately one month, with the aim to reduce the uncertainty of the statistical analysis.

In light of the comparative analysis and of the benchmark results, we are able to give a recommendation on the choice of the right mechanism.

II. RELATED WORKS

Hackenberg et al. evaluated the accuracy of RAPL measurements, in 2013 [23] and 2015 [24], using the MSR mechanism. Measurement overhead was not considered.

Huang et al. [25] showed that RAPL measurements were quite accurate for Haswell-EP processors. They assessed the performance overhead of the PAPI library. The different low-level mechanisms were not investigated.

Descrochers et al. [26] compared the values returned by RAPL for the RAM with power measurements obtained by instrumenting the hardware. Like the previous papers, this one concentrated on RAPL accuracy, not on its overhead.

In 2018, Khan et al. [27] evaluated the accuracy, update frequency and performance overhead of RAPL measurements on several parallel benchmarks. Their work focused on the MSR mechanism and did not compare it with others.

Several works were dedicated to the construction of statistical models that estimate the energy consumption of software applications [19, 18, 28]. While they compared their results with RAPL measurements, investigating the various mechanisms was out of their scope.

In 2021, Schole et al. analyzed the measurements of AMD's implementation of RAPL on the Zen 2 architecture, and revealed that it provided fewer information than Intel's [29]. One mechanism was operated, and the study concentrated on assessing the reliability of AMD's implementation.

More recently, Jay et al. [30] compared some high-level tools that internally use RAPL. They evaluated their accuracy and overhead on NAS parallel benchmarks. We also use the NAS benchmarks, and conduct a more robust statistical analysis of their results. A similar study was proposed by Huguete et al. for AI models [31]. We go deeper than both papers and analyze the underlying mechanisms instead of the higher level tools.

Due to the amount of work on RAPL measurements validation, we consider that this technology is reliable enough. Therefore, we choose not to assess again the correlation between RAPL values and external measurements. Instead, we focus on a comparative analysis of the measurement mechanisms, based on qualitative criteria as well as an experimental evaluation of the performance and energy overhead of the measurements.

III. COMPARATIVE ANALYSIS OF THE RAPL-BASED MECHANISMS

A. Operation of the RAPL interface

RAPL, that stands for Running Average Power Limit, is a power management technology first implemented by Intel in the Sandy Bridge architecture, in 2011. It allows to measure the energy consumption and to limit the power consumption of various parts of the computer, called domains. In this paper, we only look at the measurement interface, not the power limit. AMD followed suit by implementing a similar measurement interface in the first Zen architecture, in 2017. In fact, AMD's interface is so similar that it is used in the same way as Intel's. The difference lies in the addresses of the registers and in the number of available domains.

Figure 1 represents all the RAPL domains known to date. Khan et al. [27] published a similar figure, but we bring here some precisions and corrections, especially with regards to the role of *psys*.

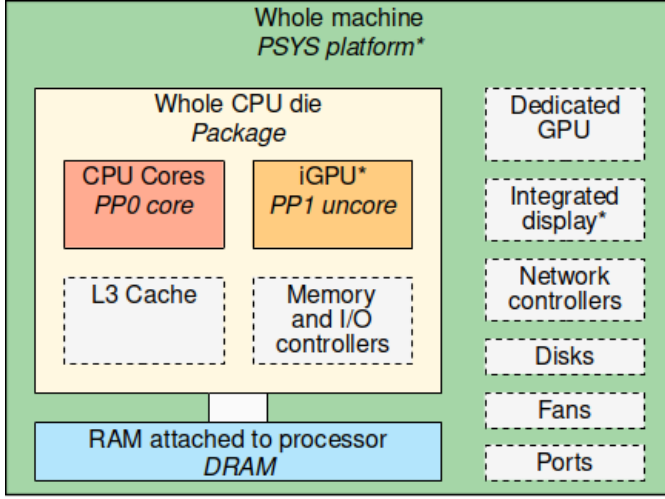


Figure 1. Hierarchy of the possible RAPL domains and their corresponding hardware components. Domain names are in italic, and grayed items do not form a domain on their own. Items marked with an asterisk are not present on servers.

First, see how the *core* and *uncore* domains are subsets of the *package* domain. This means that their reported consumption is included in the consumption of the *package* domain. Second, let us take a look at the *platform* domain, also known as *psys*. According to Intel [32], its content depends on the manufacturer of the machine. To understand its actual scope, we used two recent laptops (Lenovo Thinkpad L15 Gen1 and Alienware m17 R3) and plugged each of them into an external wattmeter. By comparing the energy consumption reported by the wattmeter with the consumption reported by the RAPL domains, we discovered that the *platform* domain reported the same consumption as an external wattmeter, that is, the total consumption of the laptop. This domain can therefore include the display, dedicated GPU and all of the other domains. Third, we want to make clear that, while the *platform* domain is unique (i.e. there is only one *platform* for the entire system-on-chip), the *dram* domain is linked to a specific CPU socket (i.e. there is one *dram* per socket). This clarifies what the Intel manual means by “directly-attached RAM”. Finally, AMD’s microarchitectures Zen 1 to Zen 4 only support the *package* and *core* domains.

Each domain has an energy counter. At regular intervals, usually every 976 microseconds [17], the energy consumed by the hardware components included in the domain since the previous increment is added to the counter. Thus, reading the counter just once is meaningless. Only the difference between two values of the counter can be interpreted as an amount of energy. Therefore, each counter needs to be read periodically. It is allowed to read the counter more often than it updates, in which case the last value is returned. Despite being possible, querying RAPL faster than its update frequency is therefore useless.

To query RAPL counters on Linux, several mechanisms can be implemented. The possible choices are described and analyzed in the following sections. Three of them are relatively well-known, yet rarely understood in depth. We also offer a new mechanism and compare it to the existing ones.

B. Comparison criteria

All the following mechanisms enable the same end result: obtaining, at a given acquisition frequency and for a given set of domains, the values of the energy consumption counters provided by the CPU’s RAPL interface. A measurement tool can be based on any of them, but that choice implies some trade-offs. In order to compare the mechanisms, we have retained the following criteria:

- the **technical difficulty** required for setting up the mechanism, i.e. the amount of time it takes to implement it, and the experience required for someone to do it.
- the amount of **knowledge** required to implement the mechanism properly, for instance whether we need to know some details of the CPU’s microarchitecture or not. It depends on the abstraction offered by the system interface.
- the **safeguards** it offers to the developer, i.e. whether it is easy to make mistakes when using the mechanism
- the **privileges** required to execute the mechanism, e.g. whether the end user of the software tool that uses the mechanism has to be `root`
- the **resiliency** of the mechanism, i.e. its ability to adapt to changes, such as the release of a new generation of processors, a new RAPL domain, etc.

C. Model Specific Registers

At the lowest level, the CPU provides model specific registers (MSR) exposing the RAPL data [17]. A user program can read them by opening `/dev/cpu/N/msr`, where *N* is the number of the core (as given by the kernel), and reading it at a specific offset. Intel and AMD use different offsets for the energy-related registers and, as far as we know, the only way to determine them is to check the processor model and to read the corresponding documentation. Note that for the MSR access to work, the `msr` kernel module has to be loaded.

The first register to read is `MSR_RAPL_POWER_UNIT`, which provides the unit of the RAPL measurements. Then, we can look at the `MSR_d_ENERGY_STATUS` registers, which contain the value of the energy counter for each domain *d*. For instance, the energy counter of the *package* domain lies in `MSR_PKG_ENERGY_STATUS`. Its value needs to be converted to Joules with the previously read unit. This conversion is not a decisive point, because it is a simple step that needs to be applied for each mechanism.

The MSR mechanism is low-level and provides no safeguard to the developer. Reading the MSR at the wrong address, or converting the bits read in the wrong way leads to a wrong measurement that can be hard to detect. Thankfully, there should be no risk of breaking the operating system nor the hardware by misusing the MSR, because we only perform reading operations, never writings. Another potential trap of

the MSR is that the energy counters overflow. These overflows occur after approximately “60 seconds under heavy load” according to Intel’s manual [17], and the only way to detect them is to poll the value frequently enough (the CPU does not signal them). It may appear as a trivial bug, however in this case it can actually be hard to detect, because the overflow frequency depends on the consumption of the machine. Testing the program on a mainstream laptop or on a server that is mostly idle can therefore make it invisible to the tester’s eyes. Many software measurement tools have been affected by bugs related to this overflow, and not all of them have resolved the issue as of this writing [33, 34, 35]. A solution to this issue is presented in section IV-B.

The kernel restricts access to the MSR because it can contain sensitive information. It has been shown that the RAPL counters could be used to perform side-channel attacks, turning into a security risk [36]. For this reason, even reading the registers requires high privileges: the binary of the measurement tool must be given the Linux capability `CAP_SYS_RAWIO` (or `CAP_SYS_ADMIN` on old kernels) or must be run as `root`. Furthermore, access control to the MSR is all-or-nothing: the `msr` module offers no way to limit a program to a restricted set of registers.

Implementing the MSR mechanism requires expert knowledge about the processor. First, one needs to know the addresses of the registers for each microarchitecture. Fortunately, CPU vendors tend to use the same addresses across successive microarchitectures. At least, a distinction between AMD and Intel has to be hardcoded. In addition, some microarchitectures have “quirks” that require a special case. For instance, some Intel processors use a fixed energy unit instead of the value of `MSR_RAPL_POWER_UNIT`. The Intel Software Developer Manual [17] gives a default value, but it does not apply when there are “quirks”.

D. Power Capping Framework

The Power Capping framework (powercap) [37] is a software interface provided by the Linux kernel on top of the low-level RAPL interface. It allows to control RAPL from userspace through the sysfs virtual file system.

The hierarchy of the sysfs under `/sys/devices/virtual/powercap/intel-rapl` resembles the hierarchy of the RAPL domains, allowing tools to discover which domains are available on the computer. An example of such a hierarchy is given by the figure 2. Each sub-folder of ‘intel-rapl’ corresponds to a domain.

In contrast to figure 1, we can see that the powercap hierarchy puts the *dram* domain inside of the *package* domain. We have found no justification of this fact in the other works related to RAPL. We think that it can be explained by the fact that, in a multi-socket system, each CPU typically has its own directly-accessible memory. Since the MSR are provided by the CPU, the *dram* energy counter of each CPU is different and only reports the consumption of the memory that is attached to it. Powercap’s authors would have chosen to reflect this by putting the *dram* domain next to the *core* domain, for each socket. Even so, our tests demonstrate that the energy

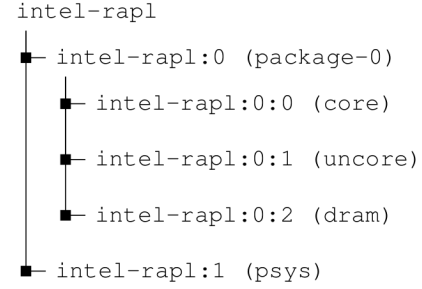


Figure 2. Folder hierarchy of the RAPL powercap sysfs on a recent laptop with an Intel CPU. For each subfolder, we indicate the name of the corresponding domain.

consumption of the memory is not included in the energy consumption of the *package*.

Being a high-level interface, powercap is easier to use than the MSR. On one hand, the measurements are provided in text files, and the MSR unit conversion is automatically applied. This is handy for testing or developing simple scripts. On the other hand, overflows frequently occur, like with the MSR mechanism. The correction to apply is slightly different because of the unit conversion.

Almost no knowledge about the processor is required, even if one must be careful with the meaning of the domains’ hierarchy, as previously explained. No special case is required, because powercap takes care of all the “quirks”.

Powercap does not require any Linux capability, it just needs read access on the `intel-rapl` directory. The file permissions or access control lists can be adjusted accordingly, or the tool can be run as `root` (the former is often preferred for security reasons).

Finally, the resiliency of the mechanism is good, thanks to an automatic adaptation of the hierarchy of the sysfs to the available domains. Of course, to benefit from this advantage, the hierarchy must not be hardcoded in the measurement tool. Our reference implementation is able to adapt to the availability of the domains.

E. perf-events

The “perf events” subsystem (hereafter “perf-events”) is another Linux kernel interface. It provides event-oriented performance monitoring capabilities [38]. There are two types of events: counting events, whose latest values must be polled, and sampling events, which are periodically added to a buffer. With a sampling event, the overflows are automatically detected. Unfortunately, the energy consumption counters have been integrated as counting events. Therefore, periodically polling the counters to detect the overflows seems to be necessary. However, overflows are extremely unlikely when using perf-events, because they are already corrected by the subsystem. Of course, the value returned by perf-events has a limited size (64 bits integer), so an overflow is still possible in theory. Even without an overflow correction, the probability of reporting erroneous measurements is much lower than with MSR and powercap.

Usually, polling the events is done from userspace. First, the list of available events can be read from the sysfs, by inspecting the `/sys/devices/power/events/` directory. Each event corresponds to a RAPL domain, and needs to be opened by calling `perf_event_open` [39]. The obtained file descriptor can then be read periodically to access the energy consumption measurements. Unlike powercap, perf-events does not organize the events in a hierarchy matching the different CPU sockets. Care must be taken to open each event exactly once for each socket, by giving the number of the first core on the socket to `perf_event_open`. Doing so requires no expert knowledge and allows to operate at a higher level of abstraction than the MSR mechanism.

In terms of resiliency, perf-events is less versatile than powercap, because one needs to call `perf_event_open`. It is therefore not available in every language, and it is much harder – though not impossible – to use it in Bash scripts. Yet, when used in a compatible environment, perf-events is as easy to maintain over time as powercap, and can easily be adapted to support new domains and microarchitectures.

It demands some privileges: either the `CAP_PERFMON` capability (since Linux 5.8, for older kernels use `CAP_SYS_ADMIN`) or the kernel setting `perf_event_paranoid` set to 0 or -1.

Another way of handling the events is to read the file descriptors from kernel space using the recent eBPF technology, as described in the following section. As far as we know, this technique has not been investigated yet.

F. perf-event via eBPF

Originally, eBPF was an abbreviation for “extended Berkeley Packet Filter”, but it is now a standalone name [40]. It refers to a technology that allows to inject code into a kernel, in particular the Linux kernel since version 3.15 [41]. Thanks to eBPF, a user program can be attached to an existing function of the kernel, which has proven to be useful for implementing fast packet filtering [42] and profiling.

In this context, we want to examine whether eBPF is a good fit for measuring the energy consumption with RAPL. The idea is that, under the hood, the aforementioned mechanisms will lead to a call to the `rdmsr x86` instruction, which needs to be done from the kernel. By reading the values with eBPF, all our measurement code would be executed by the kernel, and we would not need to switch from user mode to kernel mode. This could lower the overhead of the measurement. We test this hypothesis by designing a new measurement mechanism based on eBPF and benchmarking its implementation (see section V-A).

Figure 3 illustrates the inner workings of this new mechanism. As a preliminary step, `perf_event_open` is called and the returned file descriptors are passed to the eBPF program. This program is then injected into the Linux kernel and attached to a `SF_CPU_CLOCK` event. At a given frequency (1000 Hz in our implementation), the clock triggers the program, which reads the RAPL energy counters via perf-events and pushes the measurements into a buffer. At a regular interval managed by another timer, the userspace

program polls the content of the buffer and obtains the energy consumption measurements.

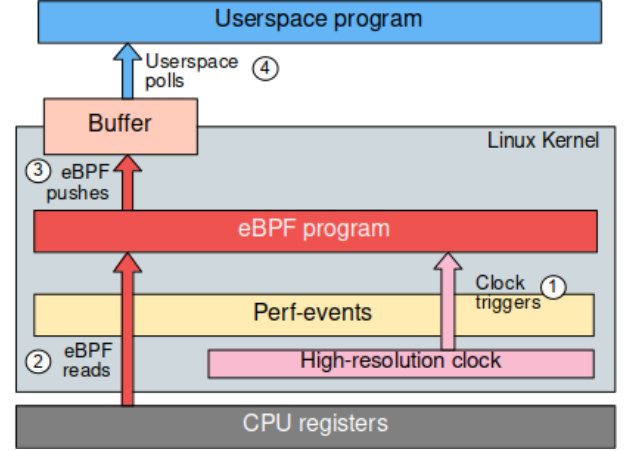


Figure 3. Measurement mechanism based on perf-events and eBPF

Using eBPF is significantly more complicated than using the “regular” userspace variant of perf-events, that is why we evaluate its technical difficulty to “high” (see section III-G).

In terms of safeguards, it is just as robust to overflows as the regular perf-events, but other types of errors can occur, especially when transferring values between user and kernel space. For example, we hit multiple difficulties related to the computation of the size of the buffers and of its elements. Our implementation is written in Rust, which prevents certain types of error. We believe that it would have been even more challenging to write it in C.

Using eBPF, for any purpose, requires to have the capability `CAP_BPF` (available since Linux 5.8, on older kernels `CAP_SYS_ADMIN` can be used instead) or to be run as root.

This mechanism demands a bit more work to adapt to changes, because the sizes of the buffers must be adjusted, and the injected code must be updated in consequence. In fact, old versions of the linux kernel are still used in production, and they do not support loops in eBPF kernel code. That is why the implementation we propose uses a match-case on the number of RAPL domains. Supporting a new domain hence requires to update the code.

G. Synthesis

The following table summarizes the results of the previous comparative analysis.

IV. IMPLEMENTATION OF A MINIMAL TOOL

A. Architecture

We implemented a minimal measurement tool with the four previously described measurement mechanisms. Its architecture is depicted on figure 4.

It features a command-line interface that allows the user to choose the RAPL domains to measure, the mechanism to use, the acquisition frequency of the polling loop, and the output file. We chose to always write the measurements during our benchmarks, in the CSV format, in order to be more

mechanism	technical difficulty	required knowledge	safeguards	privileges	resiliency
MSR	medium	CPU knowledge	none	SYS_RAWIO cap. + msr module	poor
perf-events + eBPF	high (long, complicated code)	limited	overflows unlikely, many other possible mistakes	PERFMON and BPF capabilities	manual tweaks necessary for adaptation
perf-events	low	limited	good, overflows unlikely	PERFMON capability	good
powercap	low	limited	beware of overflows	read access to one dir	good, very flexible

Table I
QUALITATIVE COMPARISON OF THE MEASUREMENT MECHANISMS

representative of a real situation. Indeed, discarding the values as soon as we read them would be useless. Thus, the impact that we evaluate in section V includes the cost of saving the measurements. To limit this cost when the acquisition frequency is high, while regularly making the data available to the user, we chose to flush the measurements to the output file once per second.

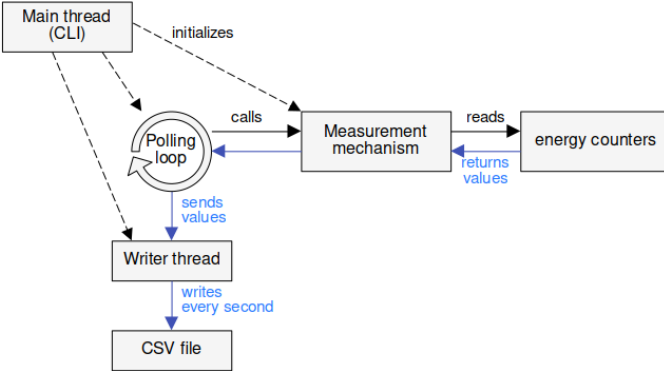


Figure 4. Architecture of our measurement tool

Compared to the existing tools, our implementation has several advantages.

First, it allows to access the “raw” energy consumption measurements, not a derived product like Code Carbon [20]. We believe that this is more rigorous and that it gives more freedom to use the measurements in a way that is suited to the context.

Second, it offers all the available mechanisms, which is useful to run experiments like the ones described in section V. Having the choice is also more robust when there is a bug in one of the mechanisms. We found that, on the AMD server used to conduct our benchmarks, powercap and perf-events did not list the same available RAPL domains. Powercap listed two domains: *package* and *core*, whereas perf-events listed all the possible domains. A manual check of each domain revealed that both lists were probably wrong, because most of the domains were unusable, and the *core* domain reported

extremely low values of about 0.001 Joule per second, which we found dubious. The perf-events sysfs has been fixed in Linux version 6.6 [43].

Finally, we took care of the previously mentioned difficulties such as overflows, and ensured that our measurement were done at the right frequency. The applied techniques are explained in the next subsections. More information is available on a public Git repository [1].

B. Correcting the overflow of the counters

As mentioned in section III-C RAPL energy measurements are prone to overflows, whose frequency depends not only on the energy consumption of the domains but also on the measurement mechanism, because they store the values in variables of different sizes and sometimes apply transformations on them. To ensure the correctness of the measurements, two conditions must be met. First, the time between two readings of the energy consumption must be smaller than the time it takes for an overflow to occur. This allows to detect an overflow between two successive measurements m_{prev} and $m_{current}$, as highlighted by previous work on RAPL. Second, the following correction must be applied:

$$\Delta m = \begin{cases} m_{current} - m_{prev} + C & \text{if } m_{current} < m_{prev} \\ m_{current} - m_{prev} & \text{otherwise} \end{cases}$$

where C is a correction constant that depends on the chosen mechanism. Table II gives the right value of C for each mechanism. Only then can Δm be used as a measure of the energy consumption of the RAPL domain during the small time period between the two measurements.

mechanism	constant C
MSR	u32::MAX i.e. 0xFFFFFFFF
perf-events	u64::MAX i.e. 0xFFFFFFFFFFFFFFFF
perf-events with eBPF	u64::MAX i.e. 0xFFFFFFFFFFFFFFFF
powercap	value given by the file <code>max_energy_uj</code> in the sysfs folder of the RAPL domain

Table II
TABLE OF THE OVERFLOW CORRECTION CONSTANT FOR EACH MEASUREMENT MECHANISM

C. Ensuring the accuracy of the acquisition frequency

As stressed previously, the acquisition frequency is a key point of a RAPL-based measurement tool. Besides the overflow correction, one could wish to increase the frequency in order to capture a more precise profile of the running applications. For example, estimating the consumption of a single function has been proven to be possible in 2012 [44]. For these reasons, it is necessary to give control of the frequency to the end user, and to make the tool precisely follow the supplied frequency.

¹<https://github.com/TheElectronWill/cpu-energy-consumption-comparative-analysis>: minimal measurement tool with an implementation of each studied RAPL-based mechanism

When we implemented our tool, we found that using a traditional `std::thread::sleep` (`nanosleep` in C) in the polling loop was not reliable enough. The variations between each sleep were big enough to be detectable, and it was impossible to reach the highest frequency of 1000 Hz. To solve this issue, we replaced the standard sleep by `timerfd`. While it cannot guarantee to perfectly respect the frequency in all conditions, especially on a non-realtime kernel, we assessed its superiority over a standard sleep.

In addition to using a more precise sleep function, we designed the tool so that the polling loop only does the minimum amount of work required to gather the measurements. Writing the data to a file is delegated to another thread. This way, the polling loop is not impacted by I/O latency.

To evaluate the effect of these two optimizations, we created three versions of the tool:

- “fully optimized” (in red): the final version of our minimal measurement tool, which includes the aforementioned optimizations (precise timer and separate I/O thread)
- “badsleep” (in blue): a version of the tool with a standard `sleep` call, and a dedicated I/O thread. This is what many existing tools use (Scaphandre, CodeCarbon, PowerAPI, etc.).
- “badsleep-st” (in green): a version of the tool with a standard `sleep` call, and a single thread (no separate I/O thread).

We executed each version with a target frequency of 1000 Hz for at least 5 minutes, and computed the actual output rate, i.e. the number of measurements per second in the result file. The first and last second were removed from the data, in order not to skew the statistics with incomplete measurement periods. Figure 5 demonstrates the benefit of our optimizations.

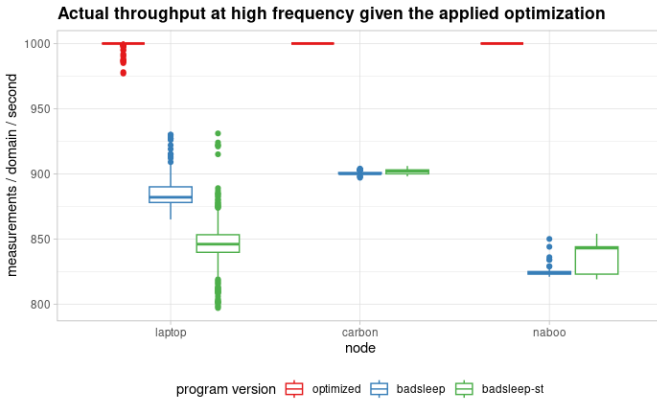


Figure 5. Boxplot of the number of measurements per domain per second, for each tested equipment (node) and version: fully optimized, without `timerfd` (`badsleep`) and without `timerfd` nor the I/O thread (`badsleep-st`)

As shown on the plot above, only the fully optimized version is able to consistently reach the target frequency of 1000 Hz. Without the two optimizations, the actual frequency decreases by 10% to 18%.

A surprising result is that moving the writing operations to another thread did not improve the output frequency on the two

tested servers. This can be explained by the fact that calling `flush` only flushes the software buffer, not the OS disk cache, which is larger on those servers than on the tested laptop. Calling `fsync` or running an I/O-intensive application in the background could make the I/O overhead more visible. Changing `std::thread::sleep` for `tokio::time::sleep` does not change the results, since the multithreaded version uses two threads with a thread pool size of two.

In any case, the experiments show that the two simple optimizations of the measurement tool have made it accurate, and allowed it to achieve a high measurement frequency. This is a clear improvement over existing RAPL-based tools, whose frequency is too often limited to a maximum of 1 or 10 Hz.

D. Brief discussion about limitations

Although it allowed us to perform extensive experiments, the minimal tool that we built is limited in some ways.

First, the tool does not support the hot-plugging of CPUs, because this feature was useless for our experiments. It could be added by using the Linux hotplug API to be notified when a CPU goes offline or online [45].

In addition, our work is strictly limited to Linux. Although the MSR should be accessible from other operating systems, we have not tried to use them. The low-level access of the registers would be similar but the higher-level interfaces, if any, would be different. To the best of our knowledge, Windows does not provide a `sysfs`-like interface to access RAPL measurements.

Finally, we have focused here on x86 Intel and AMD processors. ARM processors have not been studied because, as far as we know, most of them do not provide a RAPL-like interface. It seems to be possible on some platforms, though, since the *powermetrics* tool [46] is able to report the instantaneous power of Apple’s ARM chips. Incorporating another OS and another low-level interface into the minimal tool would require more exploratory and technical work.

V. EXPERIMENTAL STUDY OF THE MEASUREMENT OVERHEAD

A. Benchmarking Protocol

We describe here the protocol of the experimental study we conducted in order to assess the impact of the measurement mechanisms. We are interested in the performance overhead on the other running applications, but also on the energy consumption of the machine when it is idle.

1) *Test Environment*: To run the benchmarks, we used two dedicated server-class machines in Atos’ datacenters. The first one was equipped with two AMD EPYC 7702 64-cores processors and RHEL (Red-Hat Enterprise Linux) 8.7. The second was composed of two Intel Xeon E5-2680 v4 14-cores processors and ran RHEL 8.5. Both servers were running the Linux kernel version 4.18.0. On the AMD machine, the only domain available in every mechanism was the *package* domain. On the Intel machine, we were able to measure both *package* and *dram* domains.

We chose to run the well-known NAS benchmarks [47], mainly to allow our results to be compared with other papers

such as the experiments of Jay et al. [30]. In order to get various types of parallel applications, we used three NAS benchmarks: BT (block tri-diagonal solver), CG (conjugate gradient) and EP (embarrassingly parallel).

2) *Inputs and Outputs*: We recorded the running time of the NAS program, as reported by itself. Thus, this metric only includes the time it takes to solve the benchmark’s problem, not the call to `posix_spawn`, nor the final clean-up of the program’s memory, etc. We also performed some runs with a sleep of 20 minutes instead of a NAS program in order to evaluate the impact of the measurement on an idle CPU. During such a sleep, we gathered statistics about the state of the CPU with the `turbostat` command. During the experiments, we used a script to change the following variables:

- the NAS benchmark to execute (BT, CG, EP, or a 20-minutes sleep)
- the measurement mechanism to use (MSR, powercap, perf-events userspace, perf-events eBPF)
- the frequency of the measurement (0 i.e. no measurement, or 0.1Hz, 1Hz, 10Hz, 100Hz or 1000Hz)
- the RAPL domains to measure (*package only*, or *package and dram*)

That makes 96 combinations to test on the AMD server, and 192 on the Intel server. For every run of a combination, we recorded the aforementioned metrics, the variables, and the current time. At the same time, we recorded the evolution of the power consumption of each server by querying their BMC. This was done from another server, in order not to interfere with the benchmarks.

3) *Benchmark repetitions*: We ran the benchmarks for approximately one month, for as long as we could reserve the servers for our experiments. Instead of running many repetitions of one combination, then many repetitions of another combination, and so on for every combination until the end, we spread the repetitions over time. As illustrated by figure 6, we ran 3 repetitions of the first combination, then 3 repetitions of the next one, etc. Once all the combinations were done, we started all over again: 3 repetitions of the first combination (the rightmost green square on figure 6), and so on.

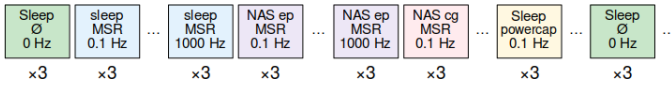


Figure 6. Benchmark repetitions spread over time

The goal of this spreading is to avoid bias we cannot control, such as another user executing something, a technical problem in the datacenter, a change in weather that affects the temperature of the room, etc. These external factors are likely to happen at a specific point in time, and thanks to this way of benchmarking they have a lower chance to spoil the results of a specific combination.

4) *Outliers*: After the execution of the benchmarks, we analyzed them with statistical tools. The first step was to eliminate the outliers in each combination. To identify them, we used the usual interquartile range. A value was considered to be an outlier if it was outside of the range $[q_1 - 3(q_3 -$

$q_1), q_3 + 3(q_3 - q_1)]$, where q_1 is the first quartile and q_3 is the third quartile. After this step, we were left with 2186 observations for the AMD server and 2564 for the Intel one.

B. Results

1) *Impact on parallel software*: Figures 7 and 8 show the running time of the three NAS benchmarks on the AMD and Intel server respectively. Our goal here is to determine whether some mechanisms are more efficient than others, and whether increasing the frequency makes the potential performance overhead more visible.

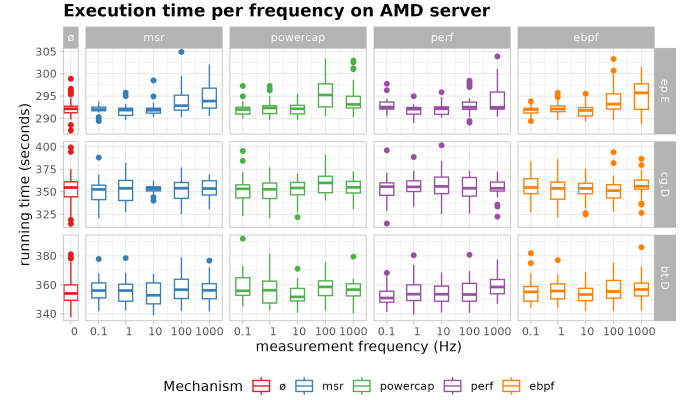


Figure 7. Performance impact of the measurement on NAS benchmarks (AMD server). The leftmost column contains the baseline.

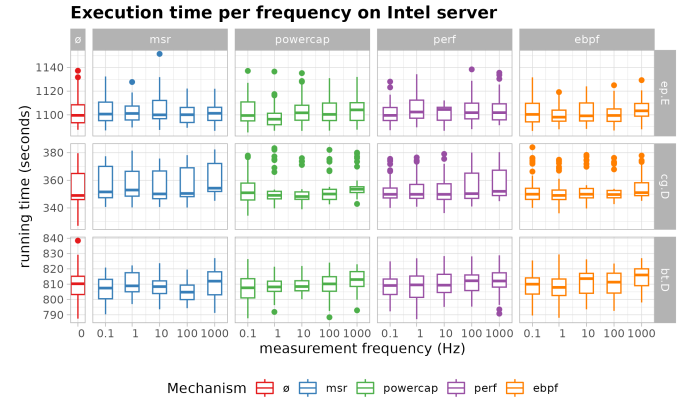


Figure 8. Performance impact of the measurement on NAS benchmarks (Intel server). The leftmost column contains the baseline.

Visually, it is difficult to see an overhead effect, even with a high measurement frequency. To test the existence of an overhead in a more rigorous way, we conducted a statistical test on each group. A group here is a subset of the data referring to the same server, NAS benchmark and RAPL mechanism. Our data did not meet the criteria for an ANOVA test, therefore we applied a one-sided Wilcoxon rank sum test. We used the standard value of 5% for the significance level α and applied the Holm-Bonferroni correction on the p-values. For each group we compared the running time of each frequency to the running time obtained without any measurement. The latter is represented on the left of the

figures, by red boxplots. Our alternative hypothesis for the statistical test is then “the running time with measurement is higher than without”.

The test shows that the overhead is statistically significant on the “ep.E” benchmark on the AMD server, at a frequency of 1000 Hz for msr, powercap and eBPF, and at a frequency of 100 Hz for powercap. While significant, the overhead remains small, with a location shift estimated at 1.5 to 3.4 seconds, that is, 0.5% to 1.2% (see appendix A). The location shift is an estimator of the median of the difference between a sample of the first group (with some measurement frequency) and a sample of the second group (without any measurement). Surprisingly, perf-events seems to have a smaller overhead than the other mechanisms on this benchmark. In particular, the low-level MSR interface seems slower than perf-events, which is counter-intuitive. We discuss this point in section V-D. Regarding the other benchmarks, there is not enough evidence to say that the measurement operation increases the running time.

The AMD server is not impacted by the measurement on all benchmarks. Only the “ep.E” benchmark, which consists of heavy floating-point operations is slowed down. The other benchmarks, which use more memory operations, are not significantly affected. Interestingly, the Intel server does not exhibit this behavior, since the only significant differences are on “cg.D” and msr, and “bt.D” and eBPF, with a location shift of respectively 5.6 seconds (1.6%) and 5.7 seconds (0.7%). An acquisition frequency of 10 Hz or less causes a negligible impact on the running time in every tested configuration.

Note that figure 8 does not make a difference between the benchmarks ran while measuring one RAPL domain and the benchmarks ran while measuring two RAPL domains. This is because, according to our analysis, that there is insufficient evidence to conclude in favor of a difference between these two cases. In other words, measuring one more domain does not change the overhead of the measurement on the benchmarks.

In addition to the running time, we recorded the power consumption of the entire servers, using their BMC. As can be seen on figures 9 and 10, we found that the measurement operation does not impact the average power when the machine is under heavy load.

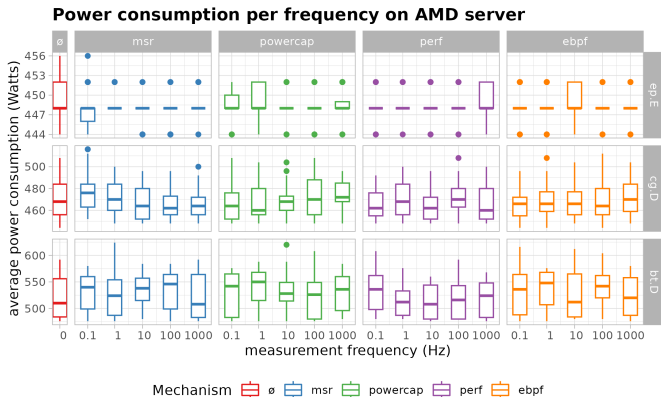


Figure 9. Impact of the measurement on power consumption during NAS benchmarks (AMD server). The leftmost column contains the baseline.

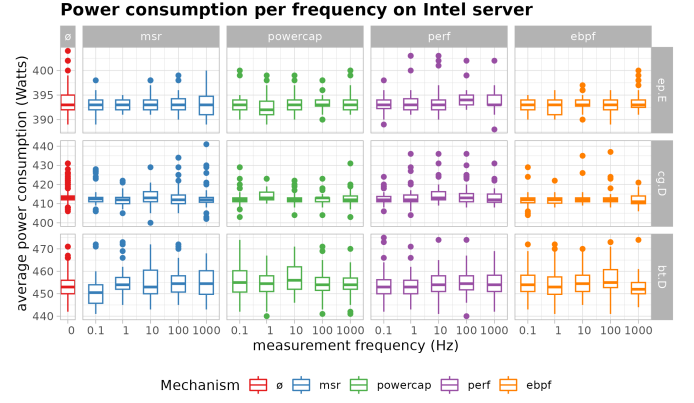


Figure 10. Impact of the measurement on power consumption during NAS benchmarks (Intel server). The leftmost column contains the baseline.

This lack of overconsumption under heavy load can be explained by the fact that, when running the benchmarks, the OS puts the CPUs at their maximum power. Hence, it cannot increase further.

2) *Impact on idle CPU:* We have just showed that the impact of RAPL energy measurements on a loaded CPU was low. In this section, we show that there can be a significant impact on an idle CPU, depending on the measurement frequency. To evaluate this impact, we used the data collected during the “sleep” runs, which consist of a simple sleep of 20 minutes. In addition to the total power read with the BMC, we collected the C-States of the processor. The state “C0” indicates that the processor is active and executing instructions. Other states are idle states, during which a subset of the CPU is disabled in order to save power.

First, let us look at the average power consumption of the server during each run of the benchmarks. As we can see in figure 11, the two servers do not react in the same way.

Average power consumption vs. frequency

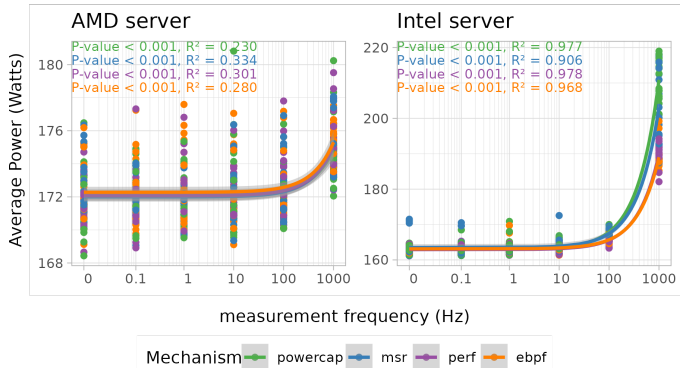


Figure 11. Relationship between acquisition frequency and idle power consumption (AMD and Intel server)

On the AMD server, the power increases starting at a measurement frequency of 1000 Hz, for which the machine draws around 2.7 to 3.3 Watts more than without measurement. On the Intel server, a Wilcoxon rank sum test indicates that the effect is significant starting at 10 Hz for powercap and

msr, and 100 Hz for perf-events and eBPF. The details are available in appendix B. At the highest frequency, the average power increases by 48.6 Watts for powercap, 40.0 for msr, 27.2 for perf-events and 25.4 for eBPF. These last two mechanisms, both based on the perf-events interface, have a clear advantage when the Intel processor is idle.

This difference between the two processors could be explained by the fact that the Intel one has more CPU states than the AMD one (note in general, Intel provides more C-States states than AMD as of 2023). As confirmed by the figure 12, querying the RAPL counters too frequently forces the Intel CPU to spend less time in C6, which consumes the least power, and more time in the higher levels, in particular C0, which consumes the most. Increasing the measurement frequency naturally causes more instructions to be executed, hence the increase of the time spent in C0. Interestingly, a high frequency of 1000 Hz makes the CPU use the intermediary states more often (C1, C1E and C3 on the Intel CPU), to the detriment of the deepest state (C6 on the Intel CPU). This means that the system determined that it was less appropriate to use the deepest state, perhaps because of its higher exit latency (it takes more time to exit C6 than to exit C3, C1E or C1). Whether Intel’s or AMD’s state management is optimal is not in the scope of this work, but it is interesting to know that using high measurement frequencies while the CPU is idle forces it to reduce its use of the deepest C-States. A good strategy would therefore be to dynamically adapt the measurement frequency to the CPU load.

CPU State versus frequency (Intel server)

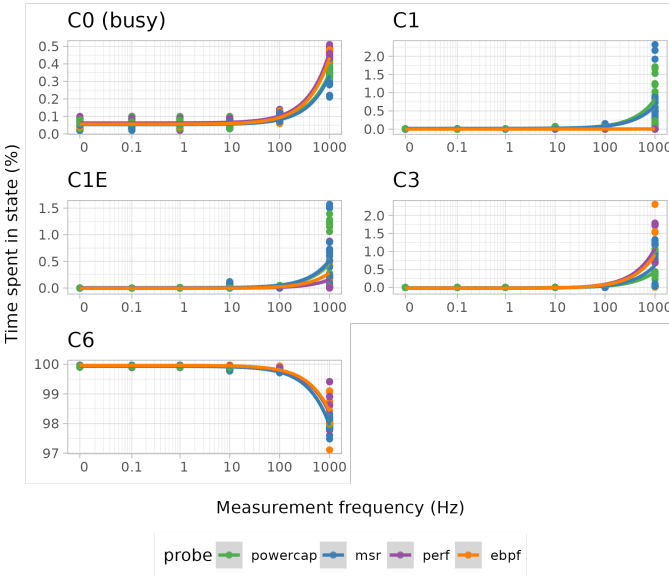


Figure 12. Relationship between acquisition frequency and idle CPU states (Intel server)

The AMD CPU also spends less time in C2 and more time in C1 and C0, as can be seen on figure 13. But there is obviously less differences between AMD C2 and C0 than Intel C6 and C0. The different mechanisms behave similarly with regards to the C-States.

CPU State versus frequency (AMD server)

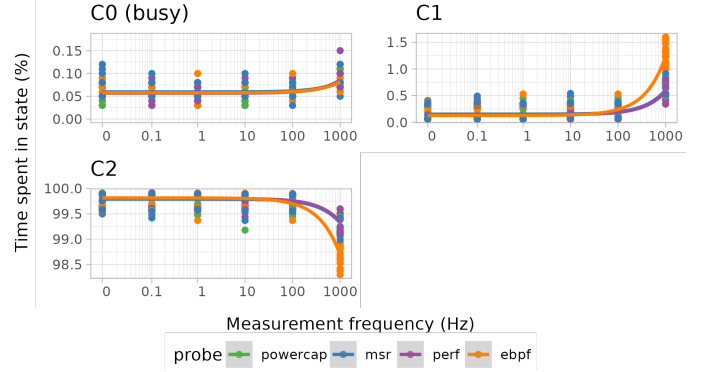


Figure 13. Relationship between acquisition frequency and idle CPU states (AMD server)

C. Measurement reading latency

The last experiment is a microbenchmark, that aims to assess the time it takes to read one value of a RAPL counter, using the various mechanisms. To build and run the microbenchmark, we used the Rust benchmarking tool *Criterion* [48] on on three machines: the two previously mentioned servers, and a Lenovo Thinkpad L15 Gen1 laptop running Ubuntu 22.04 LTS with Linux 6.1.

Table III reveals that perf-events is the fastest mechanism, followed by MSR. Not only does perf-events has the lowest latency, but it also has the lowest variance, which means that its performance is more stable than the other mechanisms. This better performance may explain the surprising result of section V-B1 on the AMD server. We see that powercap and MSR are faster on the laptop than on the servers. This could be due to the more recent kernel version of the laptop, i.e. these interfaces could have been optimized after Linux 4.18.

mechanism	time to read one RAPL counter (95% confidence interval, microseconds)		
	recent laptop	AMD server	Intel server
powercap	[1.966, 2.106]	[6.083, 6.117]	[4.230, 4.463]
perf-events	[0.5371, 0.5385]	[0.4990, 0.4993]	[0.7376, 0.7377]
msr	[0.5391, 0.5688]	[4.437, 4.448]	[2.149, 2.348]
ebpf	N/A	N/A	N/A

Table III

TABLE OF THE COST OF POLLING ONE MEASUREMENT FROM RAPL

As indicated by “N/A”s in the last row, we have chosen to exclude eBPF from this microbenchmark because of its non-representative results. In the novel eBPF-based mechanism presented in section III-F, the reading of the counter, as most of the work, happens in the kernel, and thus cannot be measured by the Criterion benchmark tool.

D. Synthesis and recommendations

Our experiments reveal that reading the RAPL energy consumption counters does not decrease the throughput of

parallel software in almost all tested configurations. A small but significant overhead has been observed on an AMD server for the cpu-intensive NAS benchmark “embarrassingly parallel”. All four tested mechanisms have a small or negligible impact on the running time of the benchmarks, even when the acquisition frequency is high. The differences reported by the microbenchmarks are less visible in the long benchmark. Compared to existing software tools, our work is significantly more lightweight. Other tools studied by prior work [30] exhibit a higher overhead (2-7%) at lower frequencies (1-10 Hz).

However, we found that measuring at high frequencies had a significant impact on the idle consumption. In particular, it reduces the time spent by the processor in low-power states, which increases the energy consumption, especially on the tested Intel processor. We hence recommend to adapt the acquisition frequency to the state of the node. Under heavy load, a high frequency can be used in order to capture more information about the running processes. On the opposite, when it is lightly loaded a lower frequency should be used. In addition, the perf-events and eBPF mechanisms seem to be the most energy-efficient when the processor is idle.

While eBPF can be used to obtain the energy consumption reported by RAPL, it has no advantage over the other mechanisms. In light of its complexity, analyzed in section III-F, we recommend not to use eBPF for this task. We make the same recommendation about MSR: unless there is no choice, that is if the OS does not provide any interface on top of the registers (like Windows), the user will prefer a higher-level mechanism.

A counter-intuitive result of our experiments is that MSR is actually slower than perf-events for energy measurement. Analyzing the implementation of the MSR module may provide an explanation. As detailed in section III-C, reading the counters is achieved by reading the file `/dev/cpu/N/msr`, where *N* is the id of the CPU core. The module relies on the file’s metadata to determine on which CPU core the `rdmsr` x86 instruction must be executed, and fetches some information related to the inode on every read. On the contrary, perf-events keeps track of the relevant information in a simple structure initialized on `perf_event_open`, not on every read.

Perf-events and powercap have similar qualities, yet important differences. On one hand, perf-events is almost immune to overflows, consumes less on idle and has a lower latency. On the other hand, powercap uses the more friendly sysfs API, which allows to get the measurements by simply reading a text file. If possible, perf-events should be preferred for its efficiency, but can be harder to use in some contexts like Bash scripts. The choice is therefore in the hands of the developer.

VI. CONCLUSION

In this work, we considered the different software mechanisms that allow to access RAPL energy counters, including a new eBPF-based mechanism that we designed. We provided a precise understanding of their operations and compared them on the basis of qualitative criteria, namely, technical difficulty, required knowledge, provided safeguards, necessary privileges

and resiliency. To highlight their differences and help the developers of software measurements tools, we released a new minimal tool with a reference implementation of each studied mechanism. We explained how difficulties such as overflows, inaccurate timing and I/O jitter could be overcome. Using this minimal tool, we conducted an experimental study on two processor models. Our results showed that, despite their differences, the mechanisms had a similar performance and energy overhead when the machine was loaded. Under nearly all benchmark configurations, running the tool had a negligible impact on the running time and power consumption, even with a high acquisition frequency of 1000 Hz. This indicates that our implementation is more lightweight and more efficient than existing software tools. It would therefore be interesting to add more features to the minimal tool, such as the estimation of the consumption of each individual process, while keeping its overhead small. The aim would then be to extend the minimal tool in order to make it better suited for end users.

We found notable differences of overhead between the mechanisms on an idle server and in microbenchmarks. In light of the experimental results and qualitative comparison, we were able to provide recommendations on the choice of the most adequate mechanism. Our main recommendation is that, quite unexpectedly, one does not need to use the most complex mechanisms (MSR, eBPF) in order to be efficient. Prefer to use the perf-events interface whenever it is possible.

Dynamically adapting the acquisition frequency to the load is also recommended. It could be particularly useful to limit the energy consumption of edge devices while collecting precise measurements. This idea could be explored in a future work, with more diverse benchmarks.

This study did not consider GPUs nor TPUs. It looks like there are fewer choices to make since GPU vendors provide a relatively high-level software library (NVML for NVidia, ROCm for AMD) and impose an underlying mechanism. Nevertheless, a natural research extension is to conduct a similar comparative and experimental study on these specialized hardware.

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APPENDIX A

RESULTS OF THE WILCOXON RANK-SUM TEST ON THE
RUNNING TIME OF THE BENCHMARKS

The *adj.pvalue* column contains the corrected p-values. The *shift* refers to the location shift estimator.

nas_bench	mechanism	freq	pvalue	adj.pvalue	shift
ep.E	msr	0.10	0.85	1.00	-0.22
ep.E	msr	1.00	0.96	1.00	-0.39
ep.E	msr	10.00	0.88	1.00	-0.29
ep.E	msr	100.00	< 0.01	0.22	1.08
ep.E	msr	1000.00	< 0.01	< 0.01	1.94
ep.E	powercap	0.10	0.80	1.00	-0.20
ep.E	powercap	1.00	0.25	1.00	0.18
ep.E	powercap	10.00	0.53	1.00	-0.02
ep.E	powercap	100.00	< 0.01	< 0.01	2.97
ep.E	powercap	1000.00	< 0.01	< 0.01	1.47
ep.E	perf-events	0.10	0.03	1.00	0.45
ep.E	perf-events	1.00	0.89	1.00	-0.25
ep.E	perf-events	10.00	0.54	1.00	-0.02
ep.E	perf-events	100.00	0.02	1.00	0.54
ep.E	perf-events	1000.00	0.01	0.51	0.75
ep.E	eBPF	0.10	0.91	1.00	-0.26
ep.E	eBPF	1.00	0.47	1.00	0.01
ep.E	eBPF	10.00	0.95	1.00	-0.48
ep.E	eBPF	100.00	< 0.01	0.10	1.26
ep.E	eBPF	1000.00	< 0.01	< 0.01	3.43
cg.D	msr	0.10	0.87	1.00	-2.24
cg.D	msr	1.00	0.44	1.00	0.50
cg.D	msr	10.00	0.76	1.00	-1.18
cg.D	msr	100.00	0.65	1.00	-0.77
cg.D	msr	1000.00	0.63	1.00	-0.78
cg.D	powercap	0.10	0.79	1.00	-1.65
cg.D	powercap	1.00	0.77	1.00	-2.01
cg.D	powercap	10.00	0.47	1.00	0.16
cg.D	powercap	100.00	0.02	0.97	5.69
cg.D	powercap	1000.00	0.39	1.00	0.45
cg.D	perf-events	0.10	0.47	1.00	0.21
cg.D	perf-events	1.00	0.28	1.00	1.22
cg.D	perf-events	10.00	0.21	1.00	2.43
cg.D	perf-events	100.00	0.45	1.00	0.55
cg.D	perf-events	1000.00	0.39	1.00	0.72
cg.D	eBPF	0.10	0.24	1.00	1.74
cg.D	eBPF	1.00	0.62	1.00	-0.73
cg.D	eBPF	10.00	0.66	1.00	-0.88
cg.D	eBPF	100.00	0.90	1.00	-3.06
cg.D	eBPF	1000.00	0.12	1.00	2.61
bt.D	msr	0.10	0.22	1.00	1.37
bt.D	msr	1.00	0.33	1.00	0.71
bt.D	msr	10.00	0.77	1.00	-1.45
bt.D	msr	100.00	0.07	1.00	2.75
bt.D	msr	1000.00	0.10	1.00	2.21
bt.D	powercap	0.10	0.04	1.00	3.02
bt.D	powercap	1.00	0.36	1.00	0.83
bt.D	powercap	10.00	0.86	1.00	-1.52
bt.D	powercap	100.00	0.04	1.00	3.04
bt.D	powercap	1000.00	0.17	1.00	1.65
bt.D	perf-events	0.10	0.96	1.00	-2.58
bt.D	perf-events	1.00	0.52	1.00	-0.09
bt.D	perf-events	10.00	0.67	1.00	-0.73
bt.D	perf-events	100.00	0.54	1.00	-0.17
bt.D	perf-events	1000.00	0.01	0.30	4.05
bt.D	eBPF	0.10	0.41	1.00	0.32
bt.D	eBPF	1.00	0.24	1.00	1.10
bt.D	eBPF	10.00	0.69	1.00	-0.76
bt.D	eBPF	100.00	0.14	1.00	1.88
bt.D	eBPF	1000.00	0.06	1.00	2.61

Table IV

WILCOXON RANK-SUM TESTS COMPARING THE RUNNING TIME WITH A MEASUREMENT AT A GIVEN FREQUENCY TO THE RUNNING TIME OF THE BENCHMARKS WITHOUT ANY MEASUREMENT (AMD SERVER).

nas_bench	mechanism	freq	pvalue	adj.pvalue	shift
ep.E	msr	0.10	0.19	1.00	1.59
ep.E	msr	1.00	0.17	1.00	1.67
ep.E	msr	10.00	0.12	1.00	2.14
ep.E	msr	100.00	0.51	1.00	-0.03
ep.E	msr	1000.00	0.26	1.00	1.13
ep.E	powercap	0.10	0.30	1.00	0.87
ep.E	powercap	1.00	0.94	1.00	-2.56
ep.E	powercap	10.00	0.13	1.00	2.09
ep.E	powercap	100.00	0.17	1.00	1.67
ep.E	powercap	1000.00	0.04	1.00	3.24
ep.E	perf-events	0.10	0.32	1.00	0.77
ep.E	perf-events	1.00	0.02	1.00	3.75
ep.E	perf-events	10.00	0.28	1.00	0.96
ep.E	perf-events	100.00	0.12	1.00	2.18
ep.E	perf-events	1000.00	0.05	1.00	3.00
ep.E	eBPF	0.10	0.15	1.00	1.98
ep.E	eBPF	1.00	0.68	1.00	-0.68
ep.E	eBPF	10.00	0.31	1.00	0.79
ep.E	eBPF	100.00	0.64	1.00	-0.56
ep.E	eBPF	1000.00	0.02	0.94	3.94
cg.D	msr	0.10	0.07	1.00	1.99
cg.D	msr	1.00	0.04	1.00	2.66
cg.D	msr	10.00	0.31	1.00	0.73
cg.D	msr	100.00	0.07	1.00	1.60
cg.D	msr	1000.00	< 0.01	< 0.01	5.58
cg.D	powercap	0.10	0.35	1.00	0.56
cg.D	powercap	1.00	0.48	1.00	0.07
cg.D	powercap	10.00	0.80	1.00	-0.97
cg.D	powercap	100.00	0.35	1.00	0.52
cg.D	powercap	1000.00	< 0.01	0.09	4.17
cg.D	perf-events	0.10	0.38	1.00	0.41
cg.D	perf-events	1.00	0.39	1.00	0.34
cg.D	perf-events	10.00	0.53	1.00	-0.16
cg.D	perf-events	100.00	0.19	1.00	1.09
cg.D	perf-events	1000.00	0.01	0.42	3.07
cg.D	eBPF	0.10	0.40	1.00	0.39
cg.D	eBPF	1.00	0.69	1.00	-0.60
cg.D	eBPF	10.00	0.36	1.00	0.44
cg.D	eBPF	100.00	0.31	1.00	0.49
cg.D	eBPF	1000.00	0.03	1.00	1.97
bt.D	msr	0.10	0.93	1.00	-2.48
bt.D	msr	1.00	0.28	1.00	0.90
bt.D	msr	10.00	0.83	1.00	-1.38
bt.D	msr	100.00	1.00	1.00	-4.38
bt.D	msr	1000.00	0.16	1.00	1.94
bt.D	powercap	0.10	0.82	1.00	-1.64
bt.D	powercap	1.00	0.81	1.00	-1.56
bt.D	powercap	10.00	0.58	1.00	-0.27
bt.D	powercap	100.00	0.41	1.00	0.35
bt.D	powercap	1000.00	0.01	0.82	3.21
bt.D	perf-events	0.10	0.69	1.00	-0.76
bt.D	perf-events	1.00	0.67	1.00	-0.87
bt.D	perf-events	10.00	0.28	1.00	0.87
bt.D	perf-events	100.00	0.12	1.00	1.89
bt.D	perf-events	1000.00	0.03	1.00	3.11
bt.D	eBPF	0.10	0.53	1.00	-0.12
bt.D	eBPF	1.00	0.84	1.00	-1.58
bt.D	eBPF	10.00	0.08	1.00	2.10
bt.D	eBPF	100.00	0.17	1.00	1.65
bt.D	eBPF	1000.00	< 0.01	0.02	5.71

Table V

WILCOXON RANK-SUM TESTS COMPARING THE RUNNING TIME WITH A MEASUREMENT AT A GIVEN FREQUENCY TO THE RUNNING TIME OF THE BENCHMARKS WITHOUT ANY MEASUREMENT (INTEL SERVER).

APPENDIX B

RESULTS OF THE WILCOXON RANK-SUM TEST ON THE
IDLE POWER CONSUMPTION

	mechanism	freq	pvalue	adj.pvalue	shift
1	powercap	0.10	0.45	1.00	0.04
2	powercap	1.00	0.64	1.00	-0.13
3	powercap	10.00	0.75	1.00	-0.24
4	powercap	100.00	0.26	1.00	0.24
5	powercap	1000.00	< 0.01	< 0.01	2.69
6	msr	0.10	0.51	1.00	-0.01
7	msr	1.00	0.91	1.00	-0.44
8	msr	10.00	0.73	1.00	-0.23
9	msr	100.00	0.17	1.00	0.30
10	msr	1000.00	< 0.01	< 0.01	2.88
11	perf-events	0.10	0.91	1.00	-0.48
12	perf-events	1.00	0.61	1.00	-0.11
13	perf-events	10.00	0.35	1.00	0.13
14	perf-events	100.00	0.10	1.00	0.46
15	perf-events	1000.00	< 0.01	< 0.01	2.77
16	eBPF	0.10	0.51	1.00	-0.01
17	eBPF	1.00	0.17	1.00	0.35
18	eBPF	10.00	0.51	1.00	-0.01
19	eBPF	100.00	0.61	1.00	-0.08
20	eBPF	1000.00	< 0.01	< 0.01	3.31

Table VI

DETAILED RESULTS OF THE WILCOXON RANK-SUM TESTS THAT COMPARES THE POWER OF THE MACHINE WITH A MEASUREMENT AT A GIVEN FREQUENCY TO THE POWER WITHOUT ANY MEASUREMENT (AMD SERVER).

	mechanism	freq	pvalue	adj.pvalue	shift
1	powercap	0.10	0.63	1.00	-0.06
2	powercap	1.00	0.14	0.95	0.27
3	powercap	10.00	< 0.01	< 0.01	1.44
4	powercap	100.00	< 0.01	< 0.01	4.63
5	powercap	1000.00	< 0.01	< 0.01	48.61
6	msr	0.10	0.67	1.00	-0.15
7	msr	1.00	0.73	1.00	-0.13
8	msr	10.00	< 0.01	< 0.01	1.64
9	msr	100.00	< 0.01	< 0.01	4.24
10	msr	1000.00	< 0.01	< 0.01	39.96
11	perf-events	0.10	0.14	0.95	0.27
12	perf-events	1.00	0.45	1.00	0.03
13	perf-events	10.00	0.06	0.56	0.35
14	perf-events	100.00	< 0.01	< 0.01	2.54
15	perf-events	1000.00	< 0.01	< 0.01	27.23
16	eBPF	0.10	0.33	1.00	0.09
17	eBPF	1.00	0.11	0.92	0.31
18	eBPF	10.00	0.05	0.47	0.44
19	eBPF	100.00	< 0.01	< 0.01	3.10
20	eBPF	1000.00	< 0.01	< 0.01	25.42

Table VII

DETAILED RESULTS OF THE WILCOXON RANK-SUM TESTS THAT COMPARES THE POWER OF THE MACHINE WITH A MEASUREMENT AT A GIVEN FREQUENCY TO THE POWER WITHOUT ANY MEASUREMENT (INTEL SERVER).